

Requirement Analysis

Functional Requirements

Functional requirements define the core actions the system must perform.

- **User Input & Request Submission:**
 - The system must provide a user-friendly form featuring a mandatory selection field for the request reason (*Lost, Replacement, or Newbie*).
 - The system must capture key employee metadata (e.g., employee name, department, and contact info) automatically upon form loading.

The screenshot shows the ServiceNow interface for an "Employee ID Card Request". The header includes the ServiceNow logo, navigation tabs (All, Favorites, History, Workspaces), and a search bar. The main content area is titled "Submit a request for a new or replacement ID card". It contains several input fields: "Department" (with "IT" entered), "Reason for Request" (with "New Hire" selected), and "Employee Name" (with "Alejandra Prenatt" entered). A "Lookup using list" button is visible next to the employee name field. On the right side, there is a sidebar with "Order this Item" (Quantity: 1), "Delivery time" (0 Days), and "Order Now" and "Add to Cart" buttons. At the bottom right, a "Shopping Cart" section shows "Empty".

- **Workflow Automation & Communication:**
 - The system must automatically generate and transmit an immediate email notification to both the requester and their direct manager upon form submission.
 - The system must evaluate approval decisions dynamically, halting downstream task generation until a manual "Approved" state is logged by an authorized manager.

Event	Notification
approval.inserted	Approval Request

Subject

Requested Item RITM0010010 Approval Request

Recipients

admin@example.com

• **Fulfillment Management:**

- Upon receiving manager approval, the system must auto-generate a structured fulfillment ticket (SCTASK) routed to the specialized badge printing or facilities team.
- The parent request ticket (RITM) must automatically transition to a "Closed Complete" state once the physical fulfillment task is manually closed by the technician.

Requested Item
RITM0010010

Number

RITM0010010

Item

Employee ID Card Request

Request

REQ0010010

Requested for

System Administrator

Due date

2026-06-16 21:51:49

Configuration item

Watch list

Opened

2026-06-16 21:51:49

Opened by

System Administrator

Stage

Request Approved

State

Open

Quantity

Pending

Open

Open

Estimated delivery

Work in Progress

Backordered

Closed Complete

Closed Incomplete

Closed Skipped

Order Guide

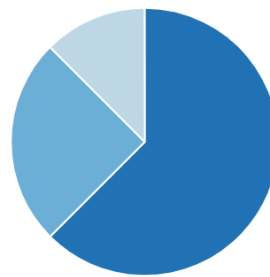
• **Data Visualization & Analytics:**

- The system must track and aggregate all card request records continuously.
- The application must provide a live visual dashboard widget (such as a donut or pie chart) that splits the metrics strictly by the user-selected reason.

Table: Requested Item [sc_req_item]

▽ All

ID Card Issue Analysis by Reason



■ Lost Card = 5 (62.5%) ■ New Hire = 2 (25%) ■ Damaged Card = 1 (12.5%)

Group by Reason for Request ▼

Non-Functional Requirements

Non-functional requirements specify the operational criteria, constraints, and quality attributes of the application.

- **Security & Access Control:**
 - Only designated line managers should have execution permissions to approve or reject the request records.
 - The logic, tables, and scripts must be entirely encapsulated within a secure, custom application namespace (Scoped Application) to prevent accidental modification or data leaks from other platform tools.



Employee ID Card Request Flow

Active

View:



TRIGGER

Service Catalog

ACTIONS

Expand Action 1 - Global - Send Email

- 1 Send Email
 - 2 Send Email
 - 3 Ask For Approval on Requested Item
 - 4 If If Approved
 - 5 then Send Email
 - 6 Create Catalog Task
- + Add an Action, Flow Logic, or Subflow

- **Scalability & Performance:**
 - The dynamic calculation of the analytics dashboard must load within acceptable platform limits (under 2 seconds) without degrading overall system performance.
 - The automation flow must trigger immediately upon form submission with near-zero latency.
- **Maintainability & Source Control:**
 - The codebase and system layout files must remain fully compatible with native platform Git integrations.
 - The package must support clean updates and commits directly to an external repository via GitHub using modern authentication (Personal Access Tokens).

3. User Personas & Core Journeys

Persona	System Role	Core User Journey
Employee (Requester)	End-User	Submits the initial form with their reasoning, receives email alerts, and checks their ticket status.
Manager (Approver)	Gatekeeper	Receives action alerts and approves or rejects the request via their approval dashboard.
Fulfiller (Facilities/Security)	Operator	Works on the generated SCTASK, prints the card, hands it over, and closes the ticket.
Administrator (Management)	Viewer	Analyzes the real-time donut chart dashboard to monitor supply costs and look for organizational trends.